

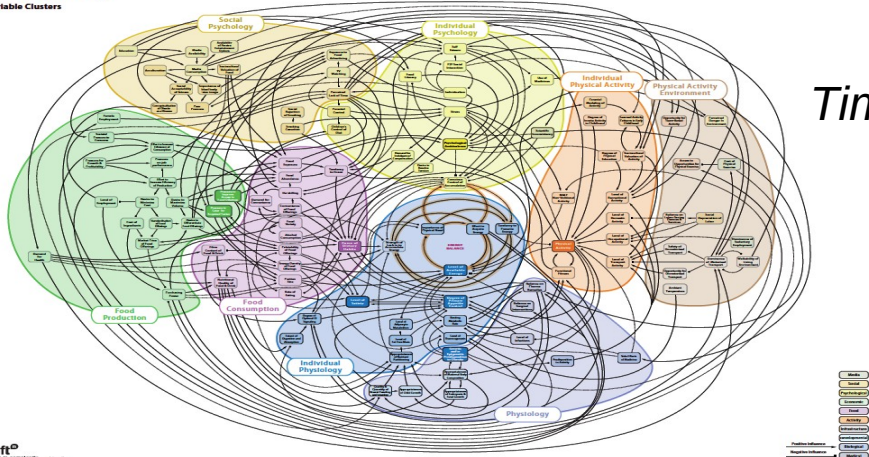
ME 2801

Introduction to Control Systems

W2: Transfer Functions

The Art of “Boxes and Arrows”

Obesity System Map
Variable Clusters



Time Domain

Physics:
Newton, Kirchhoff,
1st/2nd laws

Governing Equations (GE)
diff eqn, a set of ODEs

Transforms:
Laplace, Fourier

$\mathcal{L}\{\}$

Transfer Function (TF)

$\mathcal{L}^{-1}\{\}$

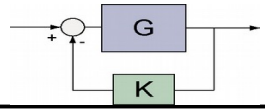
Performance and Stability

s-Domain



Time Domain

Transfer Function



A Transfer Function is the

- **ratio** of the **output** of a system to the **input** of a system,
- in the Laplace (s) domain, considering its
- **initial conditions are zero.**



So What?

- Solving differential equations (convolution), becomes multiplication.
- Enables block diagram representation.

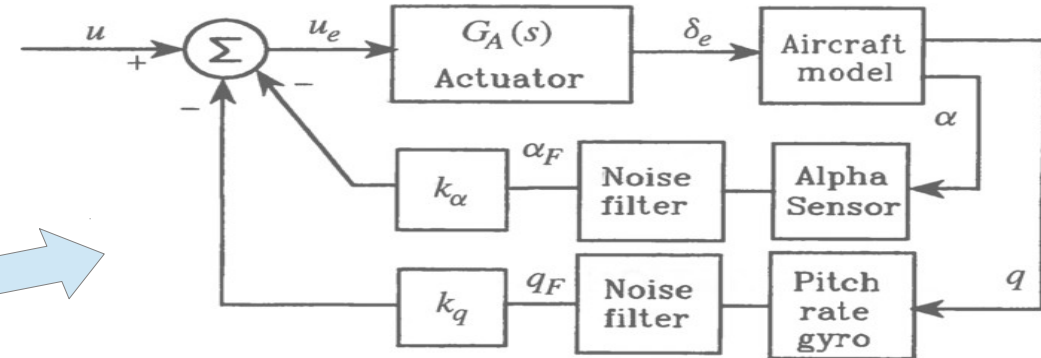
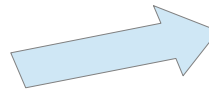


Figure 4.4-1 Pitch-axis stability augmentation.

General nth-order, LTI, differential equation

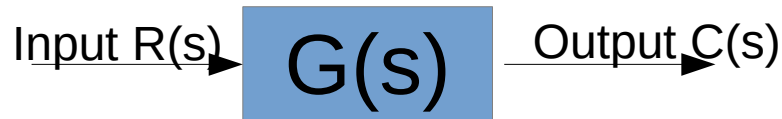
$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \dots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \dots + b_0 r(t) \quad (2.50)$$

Take Laplace transform of both side, with all **initial conditions equal zero**.

$$(a_n s^n + a_{n-1} s^{n-1} + \dots + a_0) C(s) = (b_m s^m + b_{m-1} s^{m-1} + \dots + b_0) R(s)$$

Express as ratio of output, C(s), to input, R(s).

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \dots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \dots + a_0)}$$

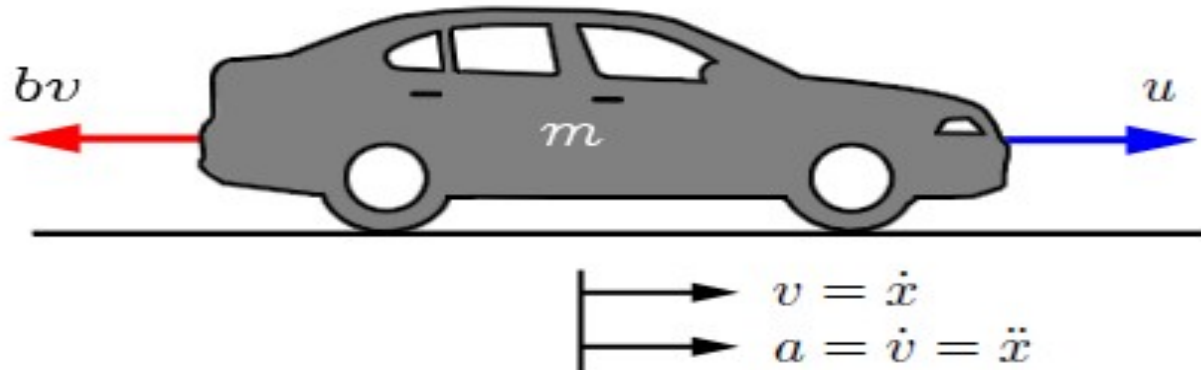


Problem Type 1:

- Given an ODE
 - Find the Transfer Function

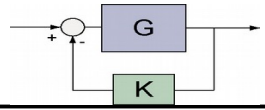
Problem Type 2:

- Given a system description (sketch, parameters, etc.)
 - Find the governing equations – typically an ODE
 - Then find the transfer function





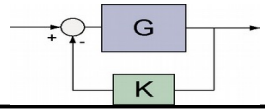
Examples – Problem Type 1



Given an ODE,
Find the transfer function...

$$m \ddot{x}(t) + b \dot{x}(t) + k x(t) = f(t)$$

$$5 \dot{y}(t) + 5 y(t) = \dot{x}(t) + 2 x(t)$$



- Ratio of polynomials: `tf(num,den)`

Properties

tf objects have the following properties:

num	Transfer function numerator coefficients. For SISO transfer functions, num is a row vector of polynomial coefficients in order of descending power (for Variable values s, z, p, or q) or in order of ascending power (for Variable values z^{-1} or q^{-1}).
den	Transfer function denominator coefficients. For SISO transfer functions, den is a row vector of polynomial coefficients in order of descending power (for Variable values s, z, p, or q) or in order of ascending power (for Variable values z^{-1} or q^{-1}).

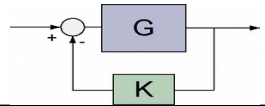
- Ratio of factors: `zpk(z,p,k)`

Properties

zpk objects have the following properties:

z	System zeros. The z property stores the transfer function zeros (the numerator roots). For SISO models, z is a vector containing the zeros. For MIMO models with Ny outputs and Nu inputs, z is a Ny-by-Nu cell array of vectors of the zeros for each input/output pair.
p	System poles. The p property stores the transfer function poles (the denominator roots). For SISO models, p is a vector containing the poles. For MIMO models with Ny outputs and Nu inputs, p is a Ny-by-Nu cell array of vectors of the poles for each input/output pair.
k	System gains. The k property stores the transfer function gains. For SISO models, k is a scalar value. For MIMO models with Ny outputs and Nu inputs, k is a Ny-by-Nu matrix storing the gains for each input/output pair.

Examples



Polynomials

Factors

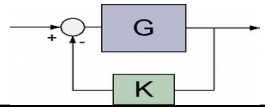
$$G(s) = \frac{5(s+1)}{(s+2)}$$

tf()

zpk()

$$H(s) = \frac{3s^2}{(s^2+4)}$$

$$K(s) = \frac{(s+1)(s+2)}{(s^2+2s+2)}$$



$$G(s) = \frac{5(s+1)}{(s+2)}$$

Polynomials

```
>> G = tf(5*[1 1],[1 2])
G =
      5 s + 5
      -----
        s + 2
```

Factors

```
>> G = zpk([-1],[-2],5)
G =
      5 (s+1)
      -----
        (s+2)
```

$$H(s) = \frac{3s^2}{(s^2+4)}$$

```
>> H = tf([3 0 0],[1 0 4])
H =
      3 s^2
      -----
      s^2 + 4
```

```
>> H = zpk([0,0],[-2i,2i],3)
H =
      3 s^2
      -----
      (s^2 + 4)
```

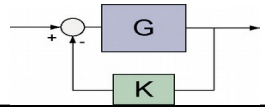
$$K(s) = \frac{(s+1)(s+2)}{(s^2+2s+2)}$$

```
>> K = tf(conv([1 1],[1 2]),[1 2 2])
K =
      s^2 + 3 s + 2
      -----
      s^2 + 2 s + 2
```

```
>> K = zpk([-1 -2],roots([1 2 2]),1)
K =
      (s+1) (s+2)
      -----
      (s^2 + 2s + 2)
```



Next Time



Modeling Introduction

- Going from physical description
 - To ODE
 - To Transfer Function